

here's a summary of the flow, results, and changes needed to run the script in different environments:

Flow:

- 1. Connect to Azure Account:** The script starts by connecting to your Azure account using managed identity authentication.
Connect-AzAccount -identity
- 2. Specify the Region:** You need to specify the region you want to check by setting the \$region variable.
\$region = "<eastus>" # Change this to your desired region
- 3. Select Subscriptions and Regions:** The script allows you to select the subscriptions and regions you want to check.
\$subscriptions = get-azsubscription | ogv -title " Select a Subscriptions to check : " -PassThru | select name, id
\$Regions = get-azlocation | ogv -title " Select a region to check : " -PassThru | select location
- 4. Loop through Subscriptions and Regions:** The script loops through the selected subscriptions and regions to get available SKUs for virtual machines in the specified region.
foreach(\$sub in \$subscriptions) {
 set-azcontext -Subscription \$(\$sub.name)
 foreach(\$Region in \$Regions) {
 \$vmSkus = Get-AzComputeResourceSku -Location \$(\$region.location)
 foreach(\$Skufamily in \$vmSkus) {
 if(\$(\$skufamily.Restrictions.reasoncode) -like '*NotAvailableForSubscription*') {
 \$reasoncode = 'Not Available For Subscription'
 } else {
 \$reasoncode = \$(\$skufamily.Restrictions.reasoncode)
 }
 \$skufamilyobj = new-object PSObject
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name Name -value
 \$(\$skufamily.name)
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name Family -value
 \$(\$skufamily.Family)
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name ResourceType -value
 \$(\$skufamily.ResourceType)
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name Location -value
 \$(\$skufamily.Locations)
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name Tier -value
 \$(\$skufamily.Tier)
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name ReasonCode -value
 "\$reasoncode"
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name Subscription -value
 \$(\$sub.name)
 \$skufamilyobj | Add-Member -MemberType NoteProperty -name "RestrictionInfo" -value
 "\$(\$skufamily.RestrictionInfo)"
 [array]\$skurestrictions += \$skufamilyobj
 }
 }

- ```

 }
}

```
5. **Generate HTML Report:** The script generates an HTML report of the SKU restrictions.  
`$CSS = @"`  
`Azure sku family restrictions : $(Get-Date -Format 'dd MMMM yyyy')`  
`"@`  
`($skurestrictions | select name, Family, ResourceType, Tier, ReasonCode, Subscription, RestrictionInfo,`  
`Location | ConvertTo-Html -Head $CSS) | Out-File "<c:\temp\restrictedSkus.html>"`  
`invoke-item "<c:\temp\restrictedSkus.html>"`
  6. **Specify Region and Subscription for Storage:** You need to specify the region and subscription for storage.  
`$Region = "<West US>" # Change this to your desired region`  
`$subscriptionselected = '<wolffentpsub>' # Change this to your desired subscription`
  7. **Export Results to CSV:** The script exports the results to a CSV file.  
`$resultsfilename = '<restrictedskus.csv>'`  
`$skurestrictions | select name, Family, ResourceType, Tier, ReasonCode, Subscription, RestrictionInfo,`  
`Location | export-csv $resultsfilename`
  8. **Storage Account Information:** You need to specify the resource group name, storage account name, and storage container name.  
`$resourcegroupname = '<wolffautomationrg>' # Change this to your resource group name`  
`$subscriptioninfo = get-azsubscription -SubscriptionName $subscriptionselected`  
`$TenantID = $subscriptioninfo | Select-Object tenantid`  
`$storageaccountname = '<wolffautosa>' # Change this to your storage account name`  
`$storagecontainer = '<restrictedskus>' # Change this to your storage container name`
  9. **Set Azure Context and Create Storage Account:** The script sets the Azure context and creates the storage account if it doesn't exist.  
`set-azcontext -Subscription $($subscriptioninfo.Name) -Tenant $($TenantID.TenantId)`  
`try {`  
 `if (!(Get-AzStorageAccount -ResourceGroupName $resourcegroupname -Name`  
`$storageaccountname)) {`  
 `New-AzStorageAccount -ResourceGroupName $resourcegroupname -Name`  
`$storageaccountname -Location $region -AccessTier Hot -SkuName Standard_LRS -Kind`  
`BlobStorage -Tag @{"owner" = "Jerry wolff"; "purpose" = "Az Automation storage write"} -Verbose`  
 `Get-AzStorageAccount -Name $storageaccountname -ResourceGroupName`  
`$resourcegroupname -verbose`  
 `}`  
`} catch {`  
 `WRITE-DEBUG "Storage Account Already Exists, Skipping Creation of $storageAccount"`  
`}`
  10. **Upload CSV to Storage Account:** The script uploads the CSV file to the specified storage account.  
`$StorageKey = (Get-AzStorageAccountKey -ResourceGroupName $resourcegroupname -`  
`StorageAccountName $storageaccountname).value | select -first 1`  
`$destContext = New-azStorageContext -StorageAccountName $storageaccountname -`  
`StorageAccountKey $StorageKey`  
`try {`  
 `if (!(get-azstoragecontainer -Name $storagecontainer -Context $destContext)) {`  
 `New-AzStorageContainer $storagecontainer -Context $destContext`  
 `}`  
`} catch {`  
 `Write-Warning " $storagecontainer container already exists"`  
`}`

```
Set-azStorageBlobContent -Container $storagecontainer -Blob $resultsfilename -File
$resultsfilename -Context $destContext -force
```

## Results:

- The script retrieves SKU restrictions for virtual machines in the specified region and generates an HTML report.
- It exports the results to a CSV file and uploads the CSV to a specified storage account.

## Changes Needed for Different Environments:

- **Region:** Change the value of \$region to your desired region.  
\$region = "<eastus>"
- **HTML Report Path:** Change the path where the HTML report will be saved.  
Out-File "<c:\temp\restrictedSkus.html>"
- **Region for Storage:** Change the value of \$Region to your desired region for storage.  
\$Region = "<West US>"
- **Subscription for Storage:** Change the value of \$subscriptionselected to your desired subscription.  
\$subscriptionselected = '<wolffentpsub>'
- **CSV Filename:** Change the value of \$resultsfilename to your desired CSV filename.  
\$resultsfilename = '<restrictedskus.csv>'
- **Resource Group Name:** Change the value of \$resourcegroupname to your resource group name.  
\$resourcegroupname = '<wolffautomationrg>'
- **Storage Account Name:** Change the value of \$storageaccountname to your storage account name.  
\$storageaccountname = '<wolffautosa>'
- **Storage Container Name:** Change the value of \$storagecontainer to your storage container name.  
\$storagecontainer = '<restrictedskus>'

## To run the PowerShell script for managing Azure resources, you need the following Azure modules:

11. **Az.Accounts:** This module is required to authenticate and manage Azure accounts.
  12. **Az.Resources:** This module is necessary for managing Azure resources, including resource groups and deployments.
  13. **Az.Compute:** This module is used to manage Azure virtual machines and related resources.
  14. **Az.Storage:** This module is required to manage Azure storage accounts and related resources.
- These modules provide the necessary cmdlets to perform the tasks outlined in the script, such as connecting to Azure, retrieving SKU restrictions, generating reports, and managing storage accounts. Make sure to install and import these modules before running the script.